

- 6 Glucagon-like peptide-1 (GLP-1) is a hormone consisting of 31 amino acids. It is secreted by specialised intestinal epithelial cells known as L cells.

Fig. 6.1 outlines the secretion of GLP-1 and its effects on the pancreas, brain and stomach.

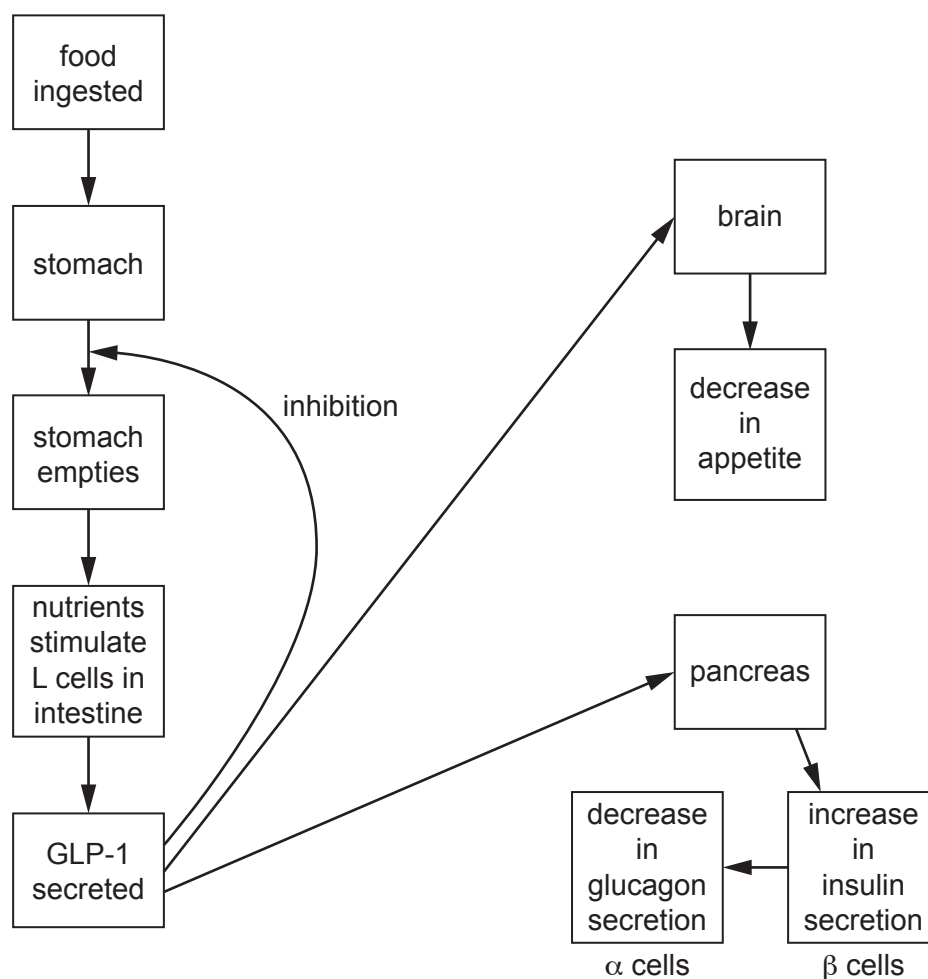


Fig. 6.1

- (a) GLP-1 binds to receptors on cell surface membranes of β cells in the pancreas. These cells secrete insulin.

- (i) Name the type of membrane component that forms a receptor in the cell surface membrane.

..... [1]

- (ii) State how GLP-1 is transported to the pancreas.

..... [1]

- (b) GLP-1 inhibits the secretion of glucagon by α cells in the pancreas.

Describe **two** processes in the liver that are stimulated by glucagon.

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..... [2]

- (c) Suggest **one** reason why GLP-1 causes a decrease in appetite.

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- (d) GLP-1 inhibits the emptying of the stomach. This is an example of negative feedback.

Explain what is meant by negative feedback.

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- (e) A transcription factor is involved in the production of GLP-1. An insertion mutation in the gene that codes for the transcription factor can affect the production of GLP-1.

Outline the effect of an insertion mutation on the production of GLP-1.

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[Total: 11]